

Speak Like A Mathematician:

corresponding parts:

congruent polygons:

congruence statement:

Congruent Polygons

Two polygons are congruent if and only if their corresponding _____ and corresponding _____ are congruent.

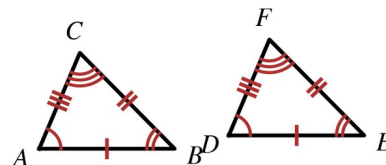
In a congruence statement the _____ of the vertices tells you which parts correspond, so $\triangle ABC \cong \triangle DEF$ means $A \leftrightarrow$ ____, $B \leftrightarrow$ ____, and $C \leftrightarrow$ ____.

Properties of Triangle Congruence: _____ ($\triangle ABC \cong \triangle ABC$), _____, and _____.

Example 1 - Name the Corresponding Parts

In the diagram, $\triangle ABC \cong \triangle DEF$. Use the congruence statement — not the picture — to answer.

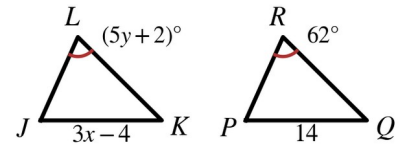
- List the three pairs of corresponding sides.
- List the three pairs of corresponding angles.
- Is it correct to write $\triangle ABC \cong \triangle EFD$? Explain.



Example 2 - Find x and y

In the diagram, $\triangle JKL \cong \triangle PQR$.

- Which side corresponds to $\overline{JK}$? Write and solve an equation for x.
- Which angle corresponds to $\angle L$? Write and solve an equation for y.
- Find JK and $m\angle L$ to check your answers.



Example 3 - Properties of Triangle Congruence

Name the property illustrated by each statement.

- $\triangle RST \cong \triangle RST$
- If $\triangle RST \cong \triangle XYZ$, then $\triangle XYZ \cong \triangle RST$.
- If $\triangle RST \cong \triangle XYZ$ and $\triangle XYZ \cong \triangle LMN$, then $\triangle RST \cong \triangle LMN$.
- Why is the Reflexive Property useful inside a proof?

Example 4 - Prove Two Triangles Congruent

This proof uses only the DEFINITION of congruent triangles, so all six pairs of parts must be shown.

Given: $\angle XWY$ and $\angle ZWY$ are right angles; $\overline{YW}$ bisects $\angle XYZ$;
W is the midpoint of $\overline{XZ}$; $\overline{XY} \cong \overline{YZ}$

Prove: $\triangle XYW \cong \triangle ZYW$

Statements	Reasons
1. $\angle XWY$ and $\angle ZWY$ are right angles	1. Given
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.
7.	7.
8.	8.
9.	9.
10.	10.

